

Aleksei Zabelin

AI Agent Engineer

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SUMMARY

I build agents that run on their own, measure what they actually do, and say so when the number is worse than it looked.

Sole engineer-owner of a SaaS platform on AWS, where I run the LLM layer myself — function-calling agents against production APIs, self-hosted retrieval, and a provider ladder that survives an outage mid-run. My last agent handled 3,271 live messages and resolved 98.3% of conversations without a human; I then found my own metric was inflated by about 20%, corrected it, and republished the lower figure. Twenty years of delivering to enterprise customers is the other half: I can sit with the department that owns the problem and work out whether an agent belongs there at all.

CORE STACK

Agents & LLM — Function calling / tool use, agent orchestration and lead state machines, versioned prompts, retrieval-augmented generation over vector stores (Supabase / Postgres + pgvector), provider routing with failover (Claude primary, OpenAI and Kimi), self-hosted model deployment (AnythingLLM on Docker), structured JSON output with intent and confidence grading

Inference & local models — Self-hosted inference on Ollama with GGUF-quantised weights, concurrency and latency benchmarking, prompt and context-window trimming, reasoning-mode ablation, cost-per-row analysis against a cloud baseline

Evaluation & measurement — Knowledge-gap tracking with open/resolved state, escalation analysis, per-message provider/model stamping for traceability, model-scored rubrics measured against a free heuristic baseline, pandas / NumPy / SciPy for result analysis, benchmarking against a stated target

Backend & Integration — Python, TypeScript, JavaScript, SQL, Bash · Node.js, Express, REST API design, webhooks, third-party API integration (amoCRM, Wazzup), OAuth / STS flows

Cloud & Delivery — AWS (EC2, S3, CloudFront, IAM, STS, WAF, Route 53), Cloudflare Workers and D1, Docker, GitHub Actions CI/CD with health-gated deploys, production monitoring and alerting, nginx, PM2, systemd

Data — PostgreSQL / Supabase, MongoDB / DocumentDB, D1, Redis job queues

AGENTS I HAVE BUILT, AND WHAT THEY MEASURED

Outbound sales agent — live campaign · May 2026 · contract work for a marketing-course business 13 active days, **3,271 messages, 471 leads**, 223 drawn into conversation. **99.5% of lead turns answered, 98.3% resolved with no human**, one escalation per 57 turns. Every turn emitted structured JSON carrying buy/stop intent, a confidence grade and a knowledge-gap flag, so the questions it could not answer surfaced as data rather than as silence. Escalations became a closing backlog — ~17–19 topics, each closed once answered. *I owned the prompt, knowledge base, intent classification, the amoCRM/Wazzup lead pipeline, paced sending, fault diagnosis and all measurement. It ran on the client's third-party orchestrator, not on my code.*

The correction. I separated "a reply went out" from "no human was needed", then found the escalation keyword had inflated the count by about 20% — the agent opened its own clarifying questions with the same phrase. I filtered it and republished the corrected number.

marketing-worker — the orchestrator rebuilt as code I own · Aug 2026 – present Cloudflare Workers and D1: lead state machine, versioned prompts, paced sending and nudge scheduling, and an **LLM provider ladder — Claude primary, OpenAI and Kimi as failover — with provider and model stamped on every message**. 75/75 tests, typecheck clean. Not yet carrying live traffic.

Function-calling agent against a production CMS API · IGOTOPIA

Runs Python against the live platform API. Non-engineers now do in conversation what previously required a developer.

Self-hosted RAG support agent · IGOTOPIA

AnythingLLM on Docker, retrieval over the company knowledge base, documents and model serving inside our own infrastructure.

Local inference spike — what the measurements actually said · Sep 2026 · own tooling Ran a scoring pipeline against a local model on Ollama with GGUF-quantised weights to test whether cloud spend was buying anything. **Per-row latency 7.5s to 2.1s** after trimming the prompt and context. Concurrency returned **+16%** and then hit a unified-memory ceiling, so it was not the answer. Turning reasoning off broke extraction, not only the rubric — a result worth knowing before shipping it. The finding that mattered: a **free ranking heuristic correlated at $r=0.536$ against the model's 0.163**, so ranking was bought from a regex and the model budget moved to the work it was actually better at.

EXPERIENCE

IGOTOPIA (Gecko Digital) — Malaysia

Mar 2022 – present · Virtual-tour SaaS for luxury hospitality on AWS Singapore

Platform & AI Engineer, Product Owner · Dec 2023 – present

Remote from Vietnam · sole engineer-owner of the Lushlist platform since Apr 2026

- **Own the platform end to end** — production and staging, four repositories, 158 commits in five months. Also product owner: I decompose the work and direct two junior developers and an operations colleague.
- **Deployed the agents above into a real business** — and decided where they did not belong.
- **Replaced edit-on-production releases with reversible ones** — built the first staging environment (isolated EC2, database, S3 prefixes, Redis) and GitHub Actions CI/CD across three applications with health-gated deploy. Wrote ~35 runbooks.
- **Built zero-cost production monitoring with guarded self-healing** — 60-second on-box sentinel, dead-man's switch, Telegram alerting. Ran it alert-only for a week, then enabled automatic recovery behind a five-layer guard stack.
- **Led a 34-finding security audit of the production API.** Every finding carried file:line evidence, a fix and a blast-radius rating. I drove the top 11 (4 critical, 6 high) into reversible fixes and closed a credential exposure spanning seven classes of secret.
- **Ran discovery, scoping and delivery for a white-labelled build** with a 3–5 person external team. Gathered the requirement, wrote the specification and milestone plan, accepted delivery.

krPano Developer & Virtual Tour Analytics · Mar 2022 – Dec 2023

- Built Python automation and data pipelines for panorama processing and publishing.
- **Cut virtual-tour load time to ~3.1s** from 4.7–7s. Benchmarked hosting and player settings against the PageSpeed Insights API, analysed the runs in pandas / NumPy / SciPy to isolate what moved Largest Contentful Paint, and tuned delivery against a 2.5s target.

Polyglot Live — polyglot-talk.ai.studio

2026 · Built for **AI Riser Vietnam 2026** — real-time speech translation for live talks: auto-detected speech in, translated audio and text out, shared to attendees' phones or a venue screen from one session link. Built on Gemini.

360° Pipeline Developer — 3DFAB Scientific & Technical Center

Dec 2019 – Jan 2022 · Saint Petersburg, Russia

360° production for **Russian Railways' staff training centres**. **Built a Python/OpenCV pre-stitching tool** that matched control points, rejected matches outside a difference threshold, and emitted a prepared PTGui project — turning a manual stitch into a review pass. *Reference available from Konstantin Konstantinov, founder and CEO.*

Freelance — 360° Photographer & krPano Developer

Sep 2015 – Dec 2019 · Saint Petersburg, Russia

Panoramic capture, stitching and tour development direct to end clients, owning each engagement from brief to published tour.

EDUCATION & CERTIFICATION

Yandex Practicum — Data Scientist programme · Mar 2022 – Dec 2023

Machine learning, statistics, data analysis and visualisation in Python.

Bonch-Bruevich Saint Petersburg State University of Telecommunications · 2000 – 2004

Diploma with Honours — Technician, Multichannel Telecommunication Systems.

Google Analytics Certification · 2024

LANGUAGES

Russian — native · English — professional working proficiency

INTERESTS

Data visualization · Photogrammetry · Embedded electronics (Arduino, ESP32, Raspberry Pi Pico) · 360° and landscape photography · Hiking and long-distance trekking · Vipassana meditation